

PokoinPOS Official Security Audit

Independent technical audit report for the PokoinPOS native blockchain node and public RPC surface.

Date: 2026-05-28

Target repository: /Users/giuseppe/pokoinpos

Public RPC endpoint: <https://rpc.pokoin.com>

Network: PokoinPoS, chainId 26062026, JSON-RPC chainId 0x18dacca

Note: Mythril is an EVM bytecode symbolic execution analyzer. The PokoinPOS native node is implemented in Go, so this audit uses Go-native security tooling plus live blockchain/RPC verification. Mythril should be used only for separate EVM smart-contract bytecode targets.

Executive Summary

The PokoinPOS native node passed the executed security and correctness checks. Unit tests, vet checks, race tests, govulncheck, and gosec completed successfully. The public RPC endpoint was reachable and returned healthy status, readiness, and the expected chain ID. No live secret/key files matching the scanned patterns were found in the repository working directory.

Area	Tool / Check	Result	Status
Source tests	go test ./...	peer, peer/account, peer/dissycrypto pass; peer/helpers has no tests	PASS
Static correctness	go vet ./...	No vet findings	PASS
Race detection	go test -race ./...	All tested packages pass; macOS linker warnings only	PASS
Vulnerability scan	govulncheck ./...	0 reachable vulnerabilities in code	PASS
Go security scan	gosec ./...	0 issues; 22 files scanned	PASS
Secret hygiene	filesystem pattern scan	No .env live, .sk, SSH key, or Firebase admin JSON files found	PASS
Live RPC	GET /health	status ok, chainHeight 791, committedHeight 790, peerCount 3	PASS
Live RPC	GET /ready	ready true, status ok	PASS
JSON-RPC	eth_chainId	0x18dacca	PASS
Mythril	EVM bytecode analyzer	Not applicable to PokoinPOS Go native node scope	N/A

Live Network Snapshot

Field	Observed value
Client version	PokoinPoS/v0.2.0
Native symbol	PKN
Health status	ok
Ready status	true
Chain height	791 at time of check
Committed height	790 at time of check
Finality depth	1
Peer count	3
Mempool depth	0

Field	Observed value
Validator stake reported by node	207730
JSON-RPC chain ID	0x18dacca

Scope and Methodology

The audit scope covers the PokoinPOS Go native node, consensus/runtime packages, local cryptographic helpers, public health/readiness endpoints, and EVM-compatible JSON-RPC compatibility surface. The audit does not assert economic guarantees, validator governance guarantees, or third-party infrastructure guarantees beyond the live endpoint checks listed above.

Tool Output Summary

```

go test ./...          PASS
go vet ./...           PASS
go test -race ./...    PASS
govulncheck ./...     PASS - 0 reachable vulnerabilities
gosec ./...            PASS - 0 issues
GET /health            PASS - status ok
GET /ready             PASS - ready true
JSON-RPC eth_chainId  PASS - 0x18dacca
Secret/key file pattern scan PASS - no matching live files found

```

Findings

ID	Severity	Finding	Status
F-01	Informational	Mythril is not applicable to the PokoinPOS Go native node because no EVM bytecode target is in scope. Use Mythril separately for EVM contracts only.	Accepted / N/A
F-02	Informational	The live network is permissioned proof-of-stake with finalityDepth 1. This is a network-design characteristic and should be documented for integrators.	Document ed

Recommendations

Maintain recurring security scans with go test, go vet, go test -race, govulncheck, and gosec before every release. Keep Go, Docker base images, and dependencies on patched versions. Continue restricting administrative endpoints and database services to private/local access. Run Mythril or Slither only for EVM smart contracts that are separately deployed or included in the integration scope.

Conclusion

Based on the executed tools and live RPC checks, the PokoinPOS native blockchain node passed the current security audit scope. No critical, high, medium, or low code-security findings were produced by the executed Go-native audit tools.